

Executive Insight Report

Online Retail Dataset | 4,338 Customers | 4 Behavioural Segments | Algorithm: K-Means (k=4)

KEY FINDING: 21% of customers (Champions) generate 67% of total revenue. The remaining 79% represent significant untapped growth — with the right segment-specific strategy.

1. PROJECT OVERVIEW

Total Customers 4,338 after cleaning & RFM build	Raw Transactions 541,909 Online Retail dataset	Optimal Clusters k = 4 Elbow + Silhouette method	Best Model K-Means Silhouette = 0.381
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An RFM (Recency, Frequency, Monetary) framework was applied to 541,909 transactions from the Online Retail dataset. After cleaning, 4,338 customers were profiled across 7 features. K-Means (k=4) was selected as the optimal model after comparing four algorithms on three cluster-quality metrics.

2. CUSTOMER SEGMENT PROFILES

Champions HIGH VALUE — PROTECT & REWARD 929 customers (21.4%) Avg. Spend: £5,745 Avg. Frequency: 10.9 orders Recency: 29 days avg. Revenue Share: 67% Most valuable segment. Frequent, high-spend, recent buyers. Prioritise retention, loyalty rewards, and early access programmes.	At-Risk WARNING — RE-ENGAGE URGENTLY 491 customers (11.3%) Avg. Spend: £829 Avg. Frequency: 2.8 orders Recency: 94 days avg. Revenue Share: 5.1% Previously active customers who have gone quiet. Win-back campaigns, personalised offers, and lapsed-customer incentives needed now.
Low-Engagement LARGEST GROUP — ACTIVATE 2,398 customers (55.3%) Avg. Spend: £698 Avg. Frequency: 2.4 orders Recency: 111 days avg. Revenue Share: 21% Majority of the base but minimal engagement. High growth potential if activated. Target with onboarding sequences, product discovery, and value nudges.	Regulars STEADY — UPSELL OPPORTUNITY 520 customers (12.0%) Avg. Spend: £1,059 Avg. Frequency: 2.5 orders Recency: 118 days avg. Revenue Share: 6.9% Consistent mid-tier buyers with high avg. basket value (£439). Strong upsell and cross-sell candidates. Worth nurturing toward Champion status.

Revenue Concentration: Champions (21% of customers) account for £5.34M — 67% of total revenue. At-Risk and Low-Engagement together represent £2.08M with significant recovery potential.

Returns Alert: Regulars carry an 88% returns ratio — an anomaly requiring investigation. This may indicate a click-and-return behaviour or fulfilment issues within this segment.

3. MODEL SELECTION

Model	Clusters	Silhouette	Davies-Bouldin	Calinski-Harabasz	Selected?
K-Means	4	0.381	1.092	3,189.7	YES
Hierarchical	4	0.301	1.252	2,882.8	—
DBSCAN	3	0.513	1.404	1,597.4	—
GMM	4	0.231	1.444	1,730.9	—

K-Means selected for its balance of inter-cluster separation (Calinski-Harabasz = 3,190) and coherent business interpretability. DBSCAN showed higher silhouette but produced only 3 clusters with poor business utility.

4. RFM FEATURE INSIGHTS

- **Recency:** Champions average 29 days vs. 111 days for Low-Engagement — a 4× gap reflecting engagement depth.
- **Frequency:** Champions average 10.9 orders; all other segments cluster below 3 — a stark loyalty divide.
- **Monetary:** Champions spend 8× more than Low-Engagement customers on average.
- **Basket Value:** Regulars have the 2nd highest avg. basket (£439) despite low frequency — upsell-ready audience.
- **Unique Products:** Champions browse 146 unique SKUs vs. 33 for Low-Engagement — breadth signals loyalty depth.
- **Weekend Ratio:** At-Risk customers show 85% weekend activity — a targeting lever for re-engagement timing.

5. ACTIONABLE RECOMMENDATIONS

#	Action	Rationale	Expected Impact
R1	Protect Champions with a VIP Programme	929 customers drive 67% of revenue. Introduce loyalty tiers, early access, and dedicated account managers to prevent churn.	Protect £5.3M revenue base
R2	Launch Re-engagement Campaign for At-Risk	491 customers averaging 94 days since last order. Deploy personalised win-back emails, time-limited discounts, and lapsed-buyer incentives immediately.	Recover ~£407K at-risk spend
R3	Activate Low-Engagement Majority	2,398 customers (55%) spend minimally. Onboarding sequences, product discovery nudges, and first-repeat-purchase offers can shift even 10% to Regulars.	+£700K+ potential uplift
R4	Upsell Regulars toward Champion Status	520 Regulars have high avg. basket value (£439) but low frequency. Cross-sell campaigns and subscription/bundle offers can increase order cadence.	Path to higher LTV
R5	Monitor Segment Migration Monthly	Re-run K-Means monthly using updated RFM features. Track customers moving between segments to measure campaign effectiveness and catch early churn signals.	Ongoing retention intelligence

6. RISKS & LIMITATIONS

- Static snapshot — RFM computed on historical data only; behaviour may have shifted seasonally.
- No demographic data — age, geography, or channel are uncontrolled; segments may mask sub-groups.
- Regulars returns ratio (88%) may reflect data quality issues or misclassified cancellations — validate before acting.
- K-Means assumes spherical clusters; non-convex customer patterns may be better served by GMM or HDBSCAN in future runs.
- Optimal k chosen by elbow + silhouette — re-validate quarterly as customer mix evolves.

7. STRATEGIC VERDICT

FOCUS: PROTECT CHAMPIONS, ACTIVATE THE MAJORITY

With 55% of customers in Low-Engagement and Champions generating 67% of revenue from just 21% of the base, the revenue opportunity is clear: even a 10% migration of Low-Engagement to Regular status adds ~£700K+ in incremental spend.

Immediate priorities: (1) lock in Champions via VIP retention, (2) re-engage At-Risk before permanent churn, (3) activate Low-Engagement with lifecycle marketing.